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(Em)Powered? Authority in a Fragmented World



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in a Multipolar World**

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Is Space the Final Frontier?

ASHWIN PRASAD

“Whoever controls low-Earth orbit controls the near-Earth space. Whoever controls near-Earth space controls Terra. Whoever controls Terra dominates the destiny of humankind.”

~ Everett C. Dolman

Historically, nations that effectively controlled the dominant strategic domain of their era gained a decisive geopolitical advantage. For the past few centuries, the arteries of global power flowed through the oceans. Command of the seas allowed nations to become empires by controlling trade, establishing colonies, projecting military strength, and amassing transcontinental wealth and influence. Today, space has emerged as this new strategic frontier. Its rise as a competitive geopolitical arena is driven by an increased reliance on its unique technologies and the nature of the domain itself.

Space – A Strategic Domain

The reliance on space is realised as national power through a suite of critical technologies providing global communication, Earth observation, and precise navigation and timing. These are technologies deployed in space that serve Earth. Satellites in orbit provide the invisible backbone for global civilian and military communications. Emerging internet

megaconstellations are rapidly expanding access and transforming communication during modern conflicts. Furthermore, Earth observation from space is used for military surveillance and resource monitoring. It underpins informed decision-making across diverse areas, including border security, maritime operations, climate action, and disaster response. Equally crucial are space-based positioning, navigation and timing services. They enable a vast array of applications—from missile strikes and commercial flights to banking transactions and smartphone GPS navigation.

This profound dependence on space technology is often unseen. Its fundamental importance makes it a potent geopolitical lever. It is critical to both economic prosperity and military effectiveness, blurring the lines between civilian and military use cases. For instance, a satellite equipped with synthetic aperture radar for weather monitoring in the Pacific can equally serve to track military vessels in the South China Sea. Similarly, a rocket designed for space launches fundamentally

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employs the same core technology as an intercontinental ballistic missile. “

The step from a V-2 type rocket to a satellite launching vehicle is a relatively small one. The same powerful motors, the same guidance systems, the same engineering know-how are involved.”

- Wernher von Braun

This dual-use nature fosters a competitive rather than cooperative dynamic in the space sector among nations. Collaboration is evident in scientific missions. The James Webb Space Telescope is a joint effort by the space agencies of the US, Europe and Canada. Cooperation thrives in such missions since the benefits of shared scientific discovery outweigh national competitive interests. This rarely extends into strategic areas. Some bilateral sale and exchange of space-based services across borders is permitted. The transfer of core technology, components, or the outsourcing of manufacturing is not. Consequently, supply chains for space are typically regionalised within the borders of spacefaring nations and their most trusted partners, splitting the world into rivalrous alliances.

In this competitive realm, the ensuing space race is not just a race between the leading spacefaring nations like the US and China. It is a gold rush to develop space-based infrastructure and control precious orbital resources. Occupying enough of an orbit is akin to blockades that deny access to adversaries—a modern analogue to controlling sea lanes. Contrary to popular belief, the usable pie of space around Earth is limited. Despite the vast distances, a few narrow rings around the Earth make for productive satellite orbits. These orbits are getting crowded. Reports from 2024 indicate that there are over 10,000 satellites orbiting Earth. Just ten years ago, it was 1,000. The carrying capacity of these orbits is limited.

Every subsequent satellite will find it harder to access the same orbits. The access to desirable orbits diminishes. It has to deal with a more congested environment. It has to invest in more complex and expensive launches and collision avoidance gear to offset the higher risk of collisions. It also has to coordinate more with other satellite operators.

Humanity's dependence on space will only increase. These conditions create an incentive for existing space powers to launch as many satellites as soon as possible. They will control the majority of the space assets and consolidate power. When developing nations begin to participate more in space activities, they will find it increasingly challenging.

“Unlike interplanetary space, Earth orbit would compare to a proximate, crowded and contestable coastline and a littoral environment, rather than a vast, remote, distant and expansive ocean.”

- Bleddyn E. Bowen

The Major Space Powers Today

Space technology, and therefore power, does not stem from a singular innovation but rather an intricate integration of multiple disciplines. These include propulsion for launch systems, material sciences for spacecraft enduring extreme speeds and heat, communication systems for antennas and data throughput, advanced cameras and sensors for Earth observation, and sophisticated software for data processing and utilisation. For that reason, a nation's ability to best leverage space for its strategic interests hinges on its proficiency across these verticals and its capacity to integrate them.

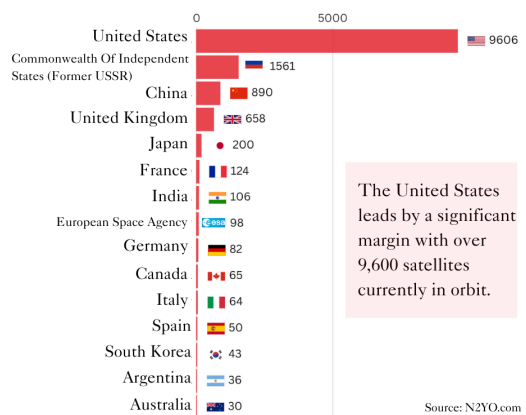
Many nations have acquired capabilities in some of these technological areas, qualifying them as spacefaring. However, only a select few have mastery over all requisite technologies and are able to integrate them to undertake space missions independently. This comprehensive, end-to-end capability is what truly confers the status of a space power on a nation.

Several factors contribute to the lead these nations maintain. Among these, indigenous space launch and transportation capabilities are critical differentiators. Fewer than a dozen entities can independently launch rockets into space. This group includes the USA, China, Russia, Japan, India, Israel, Iran, North Korea, and South Korea. Notably, only the US has operationalised reusable launch capabilities, while China, India, Russia, Europe, and Japan are at various stages of developing similar technologies. This reusability means US rockets can return after launch and be reused for subsequent missions. Reusable rockets greatly reduce launch costs and turnaround times, allowing the US to launch more frequently and extend its lead in the space race.

The tangible outcome of this end-to-end capability is the number of satellites a country deploys. Thus, the quantum of operational satellites can serve as a rough proxy for a nation's space power. By this metric, the US, China, and Russia currently own and operate the highest number of satellites. The US has the largest number of satellites, with China rapidly expanding its orbital assets in competition. Russia maintains strong space capabilities, largely stemming from its Cold War-era experience. However, sanctions have restricted its access to international trade, talent, and technology. The expansion of its operations in space lags behind that of the US and China. It is a space power in decline.

A significant portion of these national satellites serve military purposes. The number and sophistication of dedicated

Number of Satellites in Orbit by Country



military satellites reflect a nation's investment in space for defence. In contemporary military doctrine, space-based assets are central to modernisation and operational effectiveness.

Conflict in Space

"We have a saying in the space business, 'Satellites don't have mothers.' You can't hug it. You can't touch it. You can't hear it, you can't love it. It's hard for the average person to understand just how reliant their life is on space."

- John W. Raymond

This pervasive, albeit often intangible, reliance on space for military, economic, and societal functions—particularly when coupled with competitive rather than collaborative technology supply chains—makes space assets prime targets during conflicts between spacefaring nations.

Effectively, every nation on Earth is space-adjacent. Outer space, by common definition, begins at the Kármán line, merely 100 kilometres above Earth's surface. This is not a distant frontier accessible to a select few; it is a domain that directly overlays every country, making all of humanity immediate neighbours to this critical environment. As nations undertake space

activities in pursuit of their economic and security goals, space is becoming congested, contested, and competitive.

There is an advantage to be gained by sabotaging or disrupting an adversary's space assets. While the direct loss of human life from such actions might be minimal, the collapse or malfunction of space systems can have cascading and devastating consequences for a nation's infrastructure, economy, and security. Future conflicts between major powers will inevitably involve a space dimension.

Nations have pursued several avenues to develop counter-space capabilities. These include Earth-based systems such as directed energy weapons (DEWs), cyber-attacks targeting space infrastructure, and anti-satellite (ASAT) missiles. They also encompass space-based threats like co-orbital anti-satellite systems (which can manoeuvre close to and interfere with or destroy other satellites) and dedicated kinetic kill vehicles. Electronic warfare (EW) techniques, such as signal jamming and spoofing, alongside cyber-attacks, are often employed as "grey zone" tactics, falling below the threshold of overt armed conflict.

Direct-ascent anti-satellite (DA-ASAT) missiles are typically ground-launched weapons designed for "hit-to-kill" interception of satellites. The development and testing of destructive DA-ASATs date back to as early as 1959. Countries with sufficiently advanced ballistic missile programmes possess the foundational technology to develop DA-ASATs. The US, Russia, China, and India have all showcased such capabilities through destructive tests. Several other nations are believed to possess the technical prowess for latent or unproven DA-ASAT capabilities.

While non-kinetic attacks can be highly disruptive, kinetic attacks such as DA-ASATs are physically destructive—not only to their intended target but also posing a broader threat to all space assets. They generate significant orbital

debris: uncontrolled fragments travelling at hypervelocity. Collisions involving this debris can trigger a chain reaction, generating further debris and exponentially increasing the threat to operational satellites and future space missions. Ironically, the nations possessing DA-ASAT capabilities are often those with the most significant orbital assets, giving them much to lose from widespread debris proliferation. Nevertheless, an escalating conflict between space powers could lead to the widespread use of such weapons, jeopardising the space environment for all nations and potentially rendering certain orbits unusable for generations—a scenario often referred to as the Kessler Syndrome.

The issue of the weaponisation of space and its potential hazards for the global economy and security prompted the development of international space law. The cornerstone of this consensus is the Outer Space Treaty of 1967, along with its subsequent agreements and conventions. Formulated during the Cold War, its provisions reflect the geopolitical and technological context of that era. While a primary goal was to prevent the weaponisation of space (specifically, the stationing of WMDs in orbit) and promote peaceful uses, its effectiveness in addressing contemporary challenges is increasingly debated. For instance, while it prohibits national appropriation of celestial bodies, it offers limited guidance on the commercial exploitation of space resources—a rapidly emerging field. Furthermore, its enforceability is hampered by the inherent opacity of many space operations and the difficulties in attribution and verification. It primarily holds states accountable, yet the modern space landscape is increasingly characterised by the prominent role of private commercial entities, posing new regulatory dilemmas.

Private Entry into Space

The defining feature of the contemporary space era is the increasing role of private corporations in space activities. During the Cold War, the space sector—heavily influenced by national security imperatives—operated as a state-controlled endeavour managed by national space agencies. Nations maintained monopolistic control. Private players were relegated to component manufacturing. To galvanise public support, justify spending, and attract scientific talent, governments also pursued high-profile, symbolic missions to the Moon and other celestial bodies.

In the current landscape, such symbolism coexists with an increasingly substantive and indispensable reliance on space technology throughout the civilian economy and society. The proliferation of the internet and the growth of the digital economy have fuelled an insatiable demand for space-derived data and services, leading to a rapid expansion in the scale and scope of space operations.

Governments have also recognised that their military capabilities are dependent on assured access to and control over orbital assets. This, in turn, is determined by the ability to rapidly deploy, augment, or replace space infrastructure at scale. Traditional government-led agencies, often operating as monopolies, face challenges in terms of cost-efficiency and speed. There is a growing consensus that a vibrant, competitive, and decentralised industrial ecosystem is essential to meet these demands. This strategic shift—from largely secretive, state-run programmes to an emphasis on scalable, commercially driven solutions—marks the ‘third space age’.

Today, private companies routinely design, manufacture, and launch rockets, conduct Earth observation, and deploy and maintain global satellite internet megaconstellations. In response to these developments, governments worldwide are liberalising their space sectors and actively encouraging

private enterprise. National space policies, notably in countries like the US, explicitly champion this commercial-led approach. India has also recently embraced this paradigm, enacting reforms to enable greater private sector participation in its national space endeavours.

India’s Evolution in Space

India stands among the nations that made foundational investments in its space programme early in its development. Over decades, it cultivated indigenous technological capacities, achieving an independent, end-to-end spacefaring capability. Historically, this expertise resided almost exclusively within the Indian Space Research Organisation (ISRO), which functioned as the nation’s sole entity for space development and operations. Any entity seeking space-based services typically procured them from ISRO or foreign providers.

The Indian space reform was a policy response to political, economic, social, and security imperatives. Recognising that the space technology sector cannot achieve the necessary scale and strength under a single organisation, India undertook liberalisation of its space sector. This policy evolution, coinciding with the dynamic growth of the global space market, has catalysed the emergence of over 200 space start-ups within the country since the reforms were initiated. Various private Indian firms are now delivering both commercial and defence-relevant space technologies and services. Present and forthcoming national space infrastructure projects are designed to harness private sector capabilities alongside ISRO’s expertise. India’s space economy holds a 2% share of the global market, with a current valuation of USD 8.4 billion. Considering India’s space technology capabilities, this is lower than its potential. The reforms seek to leverage

private participation and growth to scale the economy to an estimated USD 44 billion by 2033.

Space technology occupies an aspirational position within Indian society, and ISRO is widely celebrated as a national asset. This broad public esteem translates into robust political will, which has historically ensured consistent funding and support for the long-term objectives of the Indian space programme. This trend will likely continue.

Space has emerged as the 21st century's paramount strategic domain. The profound reliance on space-based technologies for economic prosperity and national security, coupled with their inherent dual-use nature, has fuelled a geopolitical contest. Established space powers like the US and China vie for dominance. The development

of counter-space capabilities, alongside the limitations of existing international governance mechanisms, heightens the risk of hazard and conflict. The Indian political will and technological prowess in the space sector, and its selective legal-regulatory alignment, situate it uniquely amid the tussle between great powers for space dominance and make it a pivotal regional actor. India's space programme is both a driver and a product of global strategic competition.

"Earth is the cradle of humanity, but one cannot live in a cradle forever."

- Konstantin Tsiolkovsky ■

